JAZZ MEDIA CORES

BROADCAST MEDIA PROCESSORS

Integrated LCD/Plasma

Set-Top Box & IP Set-Top Box

Portable Video Player

Video Surveillance

Automotive or In-Flight Telematics

High performance cores that enable rapid support of robust video decoding and encoding.

The Jazz Broadcast Media Processor (BMP) cores are highly optimized solutions specifically tailored to the demanding needs of the standard-definition and high-definition digital video decoding and encoding markets. Whether you are replacing an existing solution or adding media processing to your products, the Jazz BMP cores can get you to market quickly and efficiently. They support a wide range of video, audio, image and speech standards and are optimized for the high quality video at a low cost. The BMP cores are part of the broad Jazz Media family from Improv.

New digital video devices require support for a variety of different video, audio, image and speech standards. The Jazz BMP solutions support a complete set of media software modules that can be loaded from external memory; this allows a single core to support a broad range of media standards.

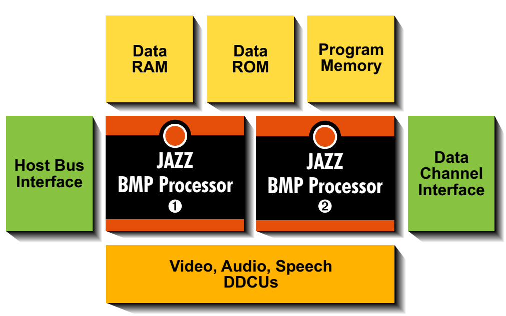
The Jazz BMP cores provide a complete package for supporting a wide variety of broadcast quality media standards in your product. The cores are pre-configured and pre-optimized and come with a complete suite of media software modules. The cores have built-in interfaces for standard on-chip buses and memory management to allow direct drop-in into an existing SoC architecture.

OPTIMIZED CORES

The Jazz BMP cores can be used as a black box implementation of complete elementary stream processing for a standard-definition or high-definition media chip design. Each core can operate as a standalone block or as a slave to a host processor.

Each BMP core has a multiple application-specific processors that work together to accelerate media processing functions. Each processor combines a general-purpose instruction set with application-specific instructions. The processors are very-long instruction word (VLIW) processors built using Improv's Jazz Composer tools.

In addition to the processor, the hardware core contains embedded memory, a host bus interface(HBI) that can connect to most standard on-chip buses including AMBA, OCP and Wishbone, and a media-aware memory interface called the Data Channel Interface (DCI).



KEY FEATURES

- Complete silicon-ready solution
- Optimized for multiple standards at a low cost
- Low clock rates for advanced media standards
- Available in standard-definition and high-definition configurations
- Fully synthesizable and foundry independent
- Supported by a comprehensive software development tool suite
- Optionally available as configurable cores for additional customization

BMP MEDIA SOFTWARE MODULES

The BMP media software modules have been designed to fully exploit the unique advantages of the BMP architecture. The BMP family includes the following modules:

- H.264 AVC Main/High Video Decoder
- MPEG-4 Advanced Simple Profile Decoder
- MPEG-4 Advanced Simple Profile Encoder
- Window Media 9 Video Decoder
- MPEG-2 MP@ML Decoder
- MPEG L1/L2 Audio Decoder
- MP3 Audio Decoder
- AAC/AAC+ Audio Decoder
- MPEG L1/L2 Audio Encoder
- MP3 Audio Encoder
- BSAC Audio Decoder
- Windows Media 9 Audio Decoder
- Dolby Digital Audio Decoder
- JPEG Still Image Codec
- G.711, G.726, G.723.1, G.729a Codecs

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Improv Systems, Inc.

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AVAILABLE PRODUCTS

The Jazz BMP family consists of the following specific products:

- **BMP Standard-Defition Decoder Core** is optimized for decoding of standard-definition (D1) video and includes all BMP decoder software modules.
- **BMP Standard-Definition Half-Duplex Codec Core** is optimized for decoding and encoding of standard-definition (D1) video and including all BMP decoder and encoder software modules.
- **BMP High-Definition Decoder Core** is optimized for decoding of high-definition video and includes all BMP decoder software modules.

IMPLEMENTATION CHARACTERISTICS

	GATE COUNT	CLOCK
BMP SD Decoder	548K gates	150 MHz
BMP SD Half-Duplex Codec	574K gates	200 MHz
BMP HD Decoder	958K gates	200 MHz

These numbers represent the maximum requirements for running the BMP Media Software modules in a TSMC 0.13um process. Video is given for D1 and HD frame sizes. In addition, 2 channel audio processing can be run on these cores at an additional 25 MHZ clock rate.

High performance is a critical component of a successful chip for these markets. The Jazz BMP cores have been highly optimized for performance both in video decoding/encoding and in memory movement on- and off-chip.

- Fully synchronous design
- Silicon-proven in 0.18um and 0.13um technologies
- Advanced power management features
- Hardware Debug Environment through JTAG port
- Support for prioritized interrupts
- Full scan with BIST for test
- Boot-loadable firmware and smart caching (Jazz Jukebox)
- No custom physical components required

DEVELOPMENT TOOLS

A complete software development environment called the Jazz Standard tool suite supports Jazzcores. This tool suite includes compiler, debugger, assembler, cycle-accurate instruction-set simulator and profiler/analyzer. The environment runs on Windows XP Professional, Windows 2000 and Windows NT environments.

The cores also come with a complete implementation environment called the Jazz Generator which creates complete synthesizable RTL, EDA tool scripts, verification environment and FPGA mapping.

As a separately purchasable item, Improv offers a development platform using Improv's FPGA-Based Jazz Rehearsal™ cards with a full verification environment to enable verification and system integration.

DELIVERABLES

- BMP Processor Core
- BMP Media Software Modules
- Verification Environment
- Jazz Standard Tool Suite
- Jazz PSA Generator
- User's Guide
- Implementation Guide
- Progammer's Guide
- Jazz Composer Tool Suite (for configurable package only)

LICENSING INFORMATION

The Jazz BMP cores are available today and can be licensed from Improv Systems. For additional information regarding the Jazz Media family and other licensable Improv technology, please contact Improv Systems at sales@improvsys.com, or visit our website at http://www.improvsys.com. A list of worldwide local representatives can be found on the website.

ABOUT IMPROV SYSTEMS, INC.

Improv Systems, Inc. develops and licenses pre-configured and configurable Jazz DSP, Jazz Media and Jazz Voice. Improv's Jazz processor is a configurable DSP that allows designers to create optimized DSP cores targeted to consumer electronics and telecommunications markets. For more information, please visit www.improvsys.com.

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